

Intra-AS Routing (OSPF)

1 The Need for Hierarchical Routing

Treating the Internet as a homogeneous collection of routers is unrealistic due to two major bottlenecks:

Challenges of Large-Scale Routing

- **Scale:** Storing and computing routing info for 100s of millions of routers is impossible. Memory overhead and broadcast traffic would saturate the network.
- **Administrative Autonomy:** ISPs want to control their own networks, run their chosen algorithms, and hide internal topology from competitors.

2 Autonomous Systems (AS)

The solution is to organize routers into **Autonomous Systems (AS)**. An AS is a group of routers under the same administrative control (typically an ISP).

Identifying an AS

- **Identity:** Each AS is identified by a globally unique **Autonomous System Number (ASN)**, assigned by ICANN registries.
- **Intra-AS Routing:** Routers *within* an AS all run the same algorithm and have full information about each other.

3 Open Shortest Path First (OSPF)

OSPF is the most widely used **intra-AS** routing protocol.

OSPF Protocol Overview

- **"Open":** Indicates the protocol specification is public (RFC 2328).
- **Algorithm:** A **Link-State (LS)** protocol using Dijkstra's algorithm to find the shortest-path tree.
- **Link Weights:** Configured by administrators (can be 1 for min-hop, or inversely proportional to link capacity).
- **Protocol Number:** Carried directly by IP (Protocol 89).

3.1 Protocol Mechanics

- **Flooding:** Broadcasts link-state info to **all** routers in the AS (not just neighbors).
- **Frequency:** Sent whenever state changes, or periodically (at least once every 30 minutes) for robustness.
- **HELLO messages:** Used to check if neighboring links are still operational.

4 Advanced OSPF Features

OSPF embodies several advanced capabilities that go beyond basic Dijkstra implementations:

Key Enhancements in OSPF

1. **Security:** Updates are **authenticated**.
 - *Simple:* Password sent in plaintext (not secure).
 - *MD5:* Uses shared secret keys and hashes for verification; prevents replay attacks.
2. **Multiple Same-Cost Paths:** Allows traffic to be balanced across multiple equal-cost paths rather than picking just one.
3. **Hierarchy Within an AS:**
 - **Areas:** AS can be split into smaller areas, each running its own OSPF.
 - **Area Border Routers (ABRs):** Responsible for routing outside their specific area.
 - **Backbone Area:** Exactly one per AS; routes traffic *between* other areas.